COURSE STRUCTURE FOR B.TECH. Mechanical Engineering FOURTH YEAR SEM – VII (w. e. f. 2024-25)

SEMESTER-VII (Subjects)				B.TECH. MECHANICAL ENGINEERING Sem-VII											
Sr. No	Category Code	Course Code	Course Name	Teaching Scheme					Exam Scheme						
									Theory			Practical		Total	
				L	T	P	C	Hrs/wk	MS	ES	CE	LE	LE/Viva	Marks	
1	PRO	24INT451	Summer Internship	0	0	0	2	0	25	25	50	-	-	100	
2	OE		Open Elective 4 (From Other Department of FoET)	3	0	0	3	3	25	25	50	-	-	100	
3	PE		Professional Elective - 3 (select from the below table)	3	0	0	3	3	25	25	50	-	-	100	
4	PE		Professional Elective - 4 (select from the below table)	3	0	0	3	3	25	25	50	-	-	100	
5	PE		Professional Elective - 5 (select from the below table)	3	0	0	3	3	25	25	50	-	-	100	
6	PCC	24ME401T	Optimization Techniques	3	0	0	3	3	25	25	50	-	-	100	
7	PCC	24ME402T	Project Management	3	0	0	3	3	25	25	50	-	-	100	
8	PCC	24ME403P	Computational Engineering Laboratory	0	0	4	2	4	-	-	-	25	25	50	
9	PRO	24PRME451	Seminar	0	0	0	1	0	-	-	-	25	25	50	
Total				**18**	**0**	**4**	**23**	**22**							

MS = Mid Semester ES = End Semester CE = Continuous Evaluation LW = Laboratory work LE = Laboratory Exam

Program Elective 3:

Course Code	Course Name
24ME431T	Exergy Analysis of Thermal Systems
24ME432T	Machine Design-II
24ME433T	Micro and Nano Manufacturing

Program Elective:4

Course Code	Course Name
24ME434T	Cryogenics
24ME435T	CAD

Program Elective 5:

Course Code	Course Name
24ME436T	Welding for Metal Joining, Surfacing and Additive Manufacturing
24ME437T	Automobile Engineering
24ME438T	Computer Aided Manufacturing
24ME439T	Non-Destructive Testing and Failure Analysis
24ME440T	Material and Procurement Management

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24ME431T					EXERGY ANALYSIS OF THERMAL SYSTEM					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	1	0	4	4	25	50	25	--	--	100

COURSE OBJECTIVES

1. To impart detailed study of energy and exergy analysis of various thermal systems
2. To study the exergy economics and Life cycle assessment

UNIT I: EXERGY FUNDAMENTALS: THERMODYNAMICS AND LAWS IN THERMAL SYSTEMS 09 Hrs.

Basic Concepts Of Thermodynamics, Thermodynamic Properties, Introduction Of Steady State And Steady Flow Properties; Application Of The First Law Of Thermodynamics For Closed / Open Systems; Application Of The Second Law Of Thermodynamics For Closed / Open Systems

UNIT II: EXERGY ANALYSIS AND THERMODYNAMIC EFFICIENCY IN THERMAL SYSTEMS 14 Hrs.

Exergy, Environment, Dead State, Limited Dead State, Definition Of Exergy Related To Kinetic And Potential Energy, Reversible Work And Irreversibility, Explanation Of The Second Law Efficiency, Thermo Mechanical Exergy, Exergy Transfer, Exergy Of A Mass, Exergy Transfer By Work And Heat; Determination Of Flow Exergy For Closed / Open Systems, Establishing Basic Exergy Equations; The Decrease Of Exergy Principle And Exergy Destruction

UNIT III: EXERGY ANALYSIS OF THERMAL SYSTEMS 09 Hrs.

Exergy Applications For Close And Open Systems, For Continuous Flow Systems Such As Turbine, Pump, Compressor, Nozzle, Heat Exchanger, Mixer, Energy Storage Systems Etc. Exergy Balance And Second Law Efficiency.

UNIT IV: THERMO-ECONOMIC APPLICATIONS OF EXERGY 10 Hrs.

Linkages Between Exergy, Economics, The Environment And Sustainability, Life Cycle Assessment: Stages Of Life Cycle Assessment, Exergetic Life Cycle Assessment, Levelized Cost Of Energy, Capital Recovery Factor, Case Studies Of Exergy Costing In Different Thermal Components

TOTAL HOURS: 42 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : **Recall** the concepts of energy, entropy and exergy
CO2 : **Apply** the first law of thermodynamics for closed / open systems.
CO3 : **Apply** the second law of thermodynamics for closed / open systems.
CO4 : **Defining** exergy analysis for open and closed systems in Engineering Thermal Systems
CO5 : **Establishing** exergy equations for each system to determine the optimum working conditions
CO6 : **Develop** the methodology for increasing the energy efficiency of thermal plants through exergy-economic analysis

TEXT/REFERENCE BOOKS

1. T. J. Kotas, **"The Exergy Method of Thermal Plant** Analysis", Elsevier
2. Ibrahim Dincer and Marc A. Roshan, **"Exergy analysis of Heating, Refrigerating and Air-conditioning",** Elsevier
3. Adrian Bejan, George Tsatsaronis, Michael J. Moran, **"Thermal Design and Optimization",** John Wiley & Sons
4. Michael J. Moran, **"Availability Analysis: A Guide to Efficient Energy Use",** Amer Society of Mechanical
5. Cengel, Y. A. (Year). **"Thermodynamics: An Engineering Approach".** McGraw-Hill

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24ME432T					Machine Design - II					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To impart knowledge on design to various force transmission elements like gears and bearings.
2. To provide the knowledge of various Industrial material handling devices.
3. To appreciate the knowledge of design of an industrial gear box.

UNIT I: DESIGN OF GEAR DRIVES 12 HRS.

Gear terminology, Kinematics of gear drive, Standard system of gear tooth, Gear materials, Design of Spur, Helical and Bevel gears.

UNIT II: DESIGN OF BEARINGS 12 HRS.

Materials and selection of bearings, Working conditions for the bearings, Hydrodynamic lubrication theory for journal bearings, Design factors, Design procedure, Design of bearings, Ball and roller bearings, Load and life rating

UNIT III: DESIGN OF GEAR BOX 6 HRS.

Geometric progression- standard step ratio, sliding mesh gear box, constant mesh gear box, synchromesh gearbox, multi speed gear box.

UNIT IV: DESIGN OF MATERIAL HANDLING EQUIPMENT 10 HRS.

Introduction of material handling equipment, Concept of material handling system design, design of ropes, Hooks.

TOTAL HOURS: 40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Design bearings that are used in various engineering and commercial applications.
CO2 : Design gears that are used in various engineering and commercial applications.
CO3 : Identify the concept of materials handling systems in industries.
CO4 : Design various components namely rope and crane hook for their safe use.
CO5 : Design gear box after understanding the concepts of various types of gear drives.
CO6 : Evaluate the various design and failure criterion as per the industry standards.

TEXT/REFERENCE BOOKS

1. MC. Sharma and K. Purohit, **"Design of Machine Elements"**, PHI Publishers.
2. U. Jindal, **"Machine Design"**, Pearson Publishers.
3. J. Shigley, C. Mischke, R. Budynas, **"Mechanical Engineering Design"**, Tata-McGraw Hill.
4. R. Norton, **"Machine Design: An Integrated Approach"**, Pearson Education Publishers.
5. V. B. Bhandari, **"Machine Design"**, Tata-McGraw Hill Publishers.

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Semester – VII

24ME433T					Micro and Nano Manufacturing					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	1	0	4	4	25	50	25	--	--	100

COURSE OBJECTIVES

1. To understand the requirements for micro/nano manufacturing
2. To introduce various mechanical micro machining techniques
3. To provide basics of various micro/nano finishing processes
4. To introduce micro/nano fabrication methods and metrology

UNIT 1 — 10 Hrs.

Introduction to Precision Engineering, Micro-electromechanical systems – micro instrumentation, micro mechatronics merits and applications. Introduction to Bulk micromachining, Surface micromachining - LIGA and SLIGA. Introduction to Micro and Nano Fabrication: basics, Photolithography and Bulk micromachining, Flowchart, basic chip making processes - – principles and applications – Examples (Field Effect Transistor). Nanofabrication using soft lithography and various approaches of soft lithography.

UNIT 2 — 10 Hrs.

Introduction to mechanical micromachining, Micro drilling, Micro turning, Micro milling and Micro grinding – process, tools and applications. Introduction to Non-conventional micro-nano manufacturing. Process, principle and applications – Micro EBM – Process principle, description and applications, Electron Beam Micro-welding – description and applications, Laser Micro welding – description and applications, Abrasive Jet Micro Machining, WAJMM, Micro EDM, Micro WEDM, Micro ECM, Micro LBM - Process principle, description and applications.

UNIT 3 — 10 Hrs.

Introduction to Micro and Nano Finishing Processes - Chemical Mechanical Polishing (CMP) – Schematic diagram, principle and applications, Magnetorheological Finishing (MRF) processes, Magnetorheological abrasive flow finishing processes (MRAFF), Force analysis of MRAFF process, Magnetorheological Jet finishing processes. Working principle and polishing performance of MR Jet Machine. Elastic Emission Machining (EEM) – machine description, applications. Ion Beam Machining (IBM) – principle, mechanism of material removal, applications.

UNIT 4 — 10 Hrs.

Introduction to Nanotechnology – Nanomaterials Classification, Carbon Nano-tubes – properties and structures, Molecular Logic Gates and Nano level Biosensors – applications. Introduction to micro and nano measurements (metrology), defining the scale, uncertainty. Optical Microscopy – description, applications and limitations. Scanning Electron Microscopy and FIB – description, principle and applications. Scanning Probe Microscopy.

Max. 40 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : **Illustrate** various mechanical micro machining methods
CO2 : **Compare** macro mechanical machining and micro mechanical machining methods
CO3 : **Understand** various micro and nano fabrication techniques
CO4 : **Judge** application of micro and nano fabrication and finishing methods
CO5 : **Identify** applications of micro and nano fabrication and finishing methods
CO6 : **Compare** different micro and nano measurement methods

TEXT/REFERENCE BOOKS

1. V.K. Jain, **"Micro-manufacturing Processes",** CRC Press, 2012
2. Nitaigour Premchand Mahalik, **"Micro-manufacturing and Nanotechnology",** 2006
3. Mark. J. Jackson, **"Micro-fabrication and Nano-manufacturing – Pulsed water drop micromachining",** CRC Press, 2006
4. Mark. J. Jackson, **"Micro and Nano-manufacturing",** Springer, 2006.
5. Lucille A. Giannuzzi, Fred A. Stevie, **"Introduction to Focused Ion Beams Instrumentation",** Theory, Techniques and Practice, Springer 2005

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Semester – VII

24ME434T					Cryogenics					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	-	-	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To impart the knowledge of production and maintaining low temperature and its application
2. To explain various cryogenic liquification system and obtain its comparative performance
3. To explain the various cry refrigerator and evaluate its performance

UNIT 1: CRYOGENICS LIQUIFICATION SYSTEMS: 11 Hrs.

Introduction to Cryogenic engineering, properties of cryogenic fluids like Oxygen, Nitrogen, Argon, Neon, Florin, Helium, Hydrogen, Safety in handling of cryogens, Properties of material at cryogenic temperature- mechanical, thermal, electrical, Super conductivity, formation of superconductivity, super conducting material behaviour, application of cryogenic systems in space, medical, industries, biological, Super conductive devices such as bearings, motors, cryotrons, magnets, D.C. transformers, space simulation,

UNIT 2: CRYOGENICS LIQUIFICATION SYSTEMS 12 Hrs.

Introduction, thermodynamically ideal systems and its performance assessment, J-T effect and analysis, inversion curve and its importance, liquefaction systems for oxygen, nitrogen, air, hydrogen, helium etc., assessment of its performance parameters, parametric analysis of the system and its effect of performance parameter, comparison of all such systems based on the performance parameters.

UNIT 3: CRYOCOOLERS AND CRYOREFRIGERATOR 12 Hrs.

Thermodynamically ideal refrigeration cycles, simple and precooled J-T refrigerator, expander-based refrigeration system, Cry coolers: Sterling refrigerator, regenerator and its effect on the performance of sterling refrigerator, Parametric study (Schmidt's Analysis), Walker's optimization charts, Design methodology of a Stirling cryocooler. single and double volume G-M refrigerator, performance comparison between sterling and G-M refrigerator, pulse tube refrigerators, adiabatic magnetization and demagnetization, magnetic refrigerator, dilution refrigerator Thermodynamically.

UNIT 4: CRYOGENIC INSULATION AND MEASUREMENT 10 Hrs.

Cryogenic insulation – expanded foams, gas filled & fibrous insulation, vacuum insulation, evacuated powder & fibrous insulation, opacified powder insulation, multilayer insulation, comparison of performance of various insulations. Dewar vessel for storage of cryogenic liquid. Cryogenic temperature, pressure, and Flow measurement

LECTURE: 45 HRS

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Recall and list the properties of cryogenic fluids and the application of cryogenics.
CO2 : Identify the behaviour of material properties at low temperature and its application.
CO3 : Understand the construction and working of various cryogenic liquification systems
CO4 : Understand the construction and working of various cryorefrigerator
CO5 : Compare and analyse the performance of various cryogenic liquification and refrigeration system.
CO6 : Analysed the various cryogenics insulations and evaluate its performance.

TEXT/REFERENCE BOOKS

1. RF Barron, "**Cryogenic systems**", Oxford University Press
2. Klaus D. Timmerhaus, Thomas M Flynn, "**Cryogenic process engineering**", Springer
3. S. S. Thipse, "**Cryogenics: A Text Book**", Alpha Science Intl Ltd
4. Graham Walker, "**Cryocooler, Part I &II**", Springer
5. M. Mukhopadhyay, "**Fundamentals of Cryogenic Engineering**", PHI

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24ME435T					Computer Aided Design					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To understand the mathematical modelling of geometric entities.
2. To apply basic knowledge of a programming software in generating algorithms for basic CAD geometric entities.
3. To apply the concepts modelling of Curves.
4. To introduce the fundamental concepts of Finite Element Analysis.

UNIT I: CAD FUNDAMENTALS AND GRAPHICS **8 Hrs.**

Introduction to CAD its benefits, CAD programming, DDA and Bresenham's Algorithm of line, mid point circle algorithm and ellipse generation algorithm.

UNIT II: GEOMETRIC TRANSFORMATIONS **9 Hrs.**

2D and 3D geometric transformations: Translation, Scaling, Rotation, Reflection, Homogeneous transformations, Projections, Orthographic and Affine transformations.

UNIT III: CURVES AND SURFACES **12 Hrs.**

Parametric representation curves: Hermit splines, Bezier curves, B-splines, NURBS, constructive solid geometry, Introduction to Surfaces, Introduction to Reverse Engineering, Introduction to Virtual prototyping, Case Studies

UNIT IV: FINITE ELEMENT METHOD **11 Hrs.**

Introduction, FEM procedure, Discretization of the Domain, Interpolation Models, Higher Order and Isoparametric Elements, Derivation of Element Matrices, Assembly of Element Matrices, Derivation of System Equations, Numerical Solution, Basic Equations and Solution Procedure, Analysis of Trusses, Beams.

TOTAL HOURS: 40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Defining the fundamental concepts related to CAD and FEM
CO2 : Interpreting the geometrical concepts on CAD entities
CO3 : Examining 2D and 3D modelling of CAD objects
CO4 : Calculating various transformations and geometrical entities
CO5 : Evaluating geometric curves and entities for given conditions
CO6 : Solving the transformations and finite element models

TEXT/REFERENCE BOOKS

1. David F Rogers and J. Adams "**Mathematical elements for computer graphic**" 2nd Ed, TMH, New Delhi.
2. Ibrahim Zeid, "**CAD/CAM Theory & Practice**", 2nd Edition, Tata Mc Graw Hill, New Delhi
3. S. S. Rao, "**The Finite Element Method in Engineering**", 5th Edition, Elsevier/Butterworth Heinemann, 2011
4. Donald Hearn, "**Computer Graphics**", 2nd Edition, Prentice-Hall, New Jersey, U.S
5. David Salomon, "**Curves and Surfaces for Computer Graphics**", Springer, New York

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Semester – VII

24ME436T					Welding for Metal Joining, Surfacing and Additive Manufacturing					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To Understand various basic and advanced welding processes and its applications
2. To learn importance of welding position, edge preparation and testing of welded joints
3. To study welding metallurgy of ferrous and non-ferrous materials
4. To analyse various cutting, cladding, hard facing and additive manufacture operations
5. To review automation and robotics systems for welding operations

UNIT 1 BASIC OF ARC WELDING PROCESSES 10 Hrs.

Importance and classification of metal joining processes. Physic of arc, power source characteristics. Metal transfer, force acting on the arc, different mode of metal transfer, heat & fluid flow in welding. SMAW, SAW, GTAW, GMAW, Basics, application, advantages and disadvantages, pulsed and synergic welding. Effect of welding parameters on welding quality.

UNIT 2 RESISTANCE, BEAM AND SOLID STATE WELDING PROCESSES 10 Hrs.

Resistance, Beam and solid state welding processes; Types of processes, Fundamental principles of operation, process characteristics, parameters and applications. Heat generation, equipment details, advantages and disadvantages. Electro slag and electro gas welding. Metallography of weld.

UNIT 3 ADVANCED WELDING AND CUTTING PROCESSES 10 Hrs.

New Development in Fusion welding processes; Activated Flux TIG process, Metal Core Arc Welding, Flux Core Arc Welding. Narrow Gap Welding processes, Hybrid Welding processes; Underwater welding & repair, cladding and surfacing. Weldability of ferrous and non-ferrous systems. Testing of welds and welding defects and significance as per various ASME/ASTM standards. Cutting Processes-Gas Cutting, Plasma cutting, Water jet cutting, Laser cutting; welding symbols and welding positions.

UNIT 4 IMPORTANCE OF CLADDING, HARD FACING , AUTOMATION AND ADDITIVE MANUFACTURING IN WELDING 10 Hrs.

Various cladding and hard facing processes, various types of automation and robotics system applicable in advanced welding processes. Importance of health and safety in various welding processes. Welding processes for additive manufacturing includes wire and arc additive manufacturing, cold metal transfer and friction stir welding for additive manufacturing and other solid state additive manufacturing. Weld 4.0, Welding simulators.

Total Hours. 40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 - Define the Various heat sources for metal joining processes.
CO2 - Explain the science behind the welding Arc, Beam, Metal transfer and Heat & Fluid flow.
CO3 - Demonstrate the various advanced welding process, applications, defects, weldability and testing.
CO4 - Analyse the various cutting, cladding methods and hard facing operations.
CO5 - Evaluate the concept of automation, robotics, Weld 4.0 and virtual welding
CO6 – Create integration of welding processes with additive manufacturing

TEXT/REFERENCE BOOKS

1. Nasir Ahmed,"**New Development in Advance Welding"**, Wood head publishing Limited, England
2. John Norrish, "**Advance Welding Processes- Technologies and Process Control"** Wood head publishing Limited, England.
3. Welding Hand **Bo**ok, "**Welding Science and Technology"**, Vol. 1, American Welding Society
4. ASM Handbook, "**Welding, Brazing and Soldering"**, Vol. 6,ASM international
5. ASM Handbook, "**Welding fundamental and Processes"** , Vol. 6A, ASM international
6. J Vora, V J Badheka, "**Advances in Welding Technologies for Process Development"**, CRC Press.
7. ASM Handbook, "**Additive Manufacturing processes",** Vol. 24, ASM international

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B. Tech. Mechanical Engineering

Semester – VII

24ME437T					Automobile Engineering					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To comprehend the function of various components of a modern-day automobile.
2. To understand the significance of automotive parameters on the performance of the automobile.
3. To outline the current standard norms and modern technologies in the domain of automobile engineering.

UNIT I: INTRODUCTION **07 Hrs.**

Introduction to Automobile and its Performance: Automobile parts, assembly of vehicle, power required for acceleration, stability of a vehicle on a slope, dynamics of a vehicle running on banked track, stability of a vehicle taking a turn. Chassis, Frame & Body: Types, Two-wheeler Chassis, comparison, Comparison of front and rear mounting of engine, aerodynamic considerations in body profiling, Ergonomic considerations, defects in frames and body.

UNIT II: TRANSMISSION SYSTEMS **12 Hrs.**

Clutch: Constructional features and working of different types of clutch (like single plate, multi plate, cone, semi– centrifugal, fully, centrifugal, wet etc.) used in automobiles, calculation of surface area and number of driving and driven plates, fluid coupling. Gear Box: Functions of gearbox, need of gear box, gears & gear ratios, principle of gearing, types of gear boxes, manual gearboxes. Automatic Transmission: Basic devices used in automatic transmission, principle of epicyclic gearing, torque converter, free wheel clutch, over speed drive and its working, semi, fully automatic transmission, continuously variable transmission (CVT), Adaptive 4-wheel Drive System. Propeller Shaft: Propeller shafts and their types, fluid drive and fluid flywheel, universal joints, hotch-kiss drive, torque tube drive. Differential: Principle of the differential, locking differential, limited slip differential Final Drive and Rear Axle: Final drives and its types, hypoid type final drive, rear axle, rear axle drives, rear axle shaft supporting, rear axle casing, axle breather, oil retention

UNIT III: AUTOMOTIVE SUB-ASSEMBLIES **11 Hrs.**

Front Axle: Types, construction, components and their functions. Suspension System: Principle, type of suspension system, conventional and independent front and rear axle, spring, rubber and air suspensions, automatic, hydro suspension system, shock absorbers. Semi-Active and Fully Active Suspension System. Steering System: Steering layout, types of steering gears, steering linkages, steering mechanism, steering geometry, measurement and adjustment of various steering system layouts, steering ratio, under steering and over steering, power assisted steering, wheel alignment. Brakes: Principle, braking distance, braking efficiency, weight transfer, wheel skidding, principle and working of various types of brakes, anti–lock brake systems (ABS). Battery: Construction, Types of batteries, working, methods of rating, faults, charging methods. Lighting system: Wiring system, head lights, indicating lights. Accessories and Safety systems like air bags, safety belts

UNIT IV: REGULATION AND MODERN VEHICLES **10 Hrs.**

Regulation and Standardization of Vehicles: Motor vehicle act, registration of motor vehicles, driving license, control of traffic, insurance against third party, traffic signs, central motor vehicle rules, vehicle safety standards and regulations, vehicle Pollution Norms. Modern Vehicles: Construction and operational features of four wheelers available in Indian market, Vehicle Management Systems, ESP, Hill Assist, ACC, EBD, Lane Assist, Smart Features. Introduction to electric vehicles & hybrid vehicles. Connected Cars and IOT, ADAS and Autonomous Vehicles.

TOTAL HOURS: 10 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Analyse the performance characteristics of an Automobile and the factors affecting it.
CO2 : Analyse the transmission system of an Automobile.
CO3 : Explain the working of suspension and steering systems in an Automobile.
CO4 : Compare the relative working and performance of various auxiliaries and sub-systems of an Automobile.
CO5 : Summarize the function of electrical and safety systems in an Automobile.
CO6 : Identify the standard norms and modern technologies in the domain of Automobile Engineering.

TEXT/REFERENCE BOOKS

1. Kirpal Singh, "**Automobile Engineering Vol-I & II"**, Standard Pub. & Distributer.
2. W. Crouse, "**Automotive Mechanics"**, Tata McGraw Hill.
3. S. K. Gupta, "**A Textbook of Automobile Engineering"**, S. Chand Pub.
4. K. K. Jain and R. B. Asthana, "**Automobile Engineering",** McGraw Hill Pub.

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Semester – VII

24ME438T					Computer Aided Manufacturing					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To understand detail of CNC machines
2. To implement concept of manual part programming
3. To provide fundamentals of flexible manufacturing system and robotics
4. To understand group technology and cellular manufacturing

UNIT I INTRODUCTION TO CNC MACHINES 10 Hrs.

Introduction to CAM, Nature & type of manufacturing system, NC, CNC and DNC systems, Elements of CNC machine - Slideways, motion transmission elements, Automatic tool changers and multiple pallet systems, feedback devices - encoders and transducers, sensors, actuators, Spindle drives and axes drives, Tooling for CNC machines-Tool preset and qualified tools, work and tool holding devices.

UNIT II: CNC PART PROGRAMMING 10 Hrs.

Axis identification and coordinate systems, Programming formats. Manual part programming for a turning center- Do loop, Sub routines and canned CNC lathe cycles. Manual part programming for CNC milling center - Do loop, Subroutines and Canned milling cycles. CNC part programming using APT language.

UNIT III: FLEXIBLE MANUFACTURING SYSTEM 10 Hrs.

Introduction of FMS, Types of FMS, Manufacturing Cells, JIT & GT applied to FMS, FMC, Tool Management, industrial robotics and material Handling, AS/RS, AGVS , RGV, Flexible Fixturing, Flexible Assembly Systems, FMS scheduling, sequencing, FMS lay out and essentials

UNIT IV: GROUP TECHNOLOGY AND CELLULAR MANUFACTURING 10 Hrs.

Introduction, part families, part classification and coding, machining cells, production flow analysis, Cellular Manufacturing, Computer Aided Production management- Introduction, PPC fundamentals, Problems with traditional PPC, use of computer in PPC such as CAPP, MRPI, MRPII, CAGC etc.

TOTAL HOURS: 40 Hrs.

COURSE OUTCOMES

On completion of the course, students will be able to:

CO1 : Understand CNC machine structures and system drives
CO2 : Analyze basics and advancement in NC and CNC for automatic manufacturing
CO3 : Create manual and APT part programs for complex profiles for Lath and milling machine
CO4 : Understand role of FMS, JIT, and Robotics in material movement in manufacturing
CO5 : Apply the concepts of group technology and cellular manufacturing in automation
CO6 : Understand computer aided production management in industries

TEXT/REFERENCE BOOKS

1. Chang, Tien-Chien, and Richard A. Wysk. "**Computer-aided manufacturing**", Prentice Hall PTR
2. Groover, Mikell P. "**Automation, production systems, and computer-integrated manufacturing"** Pearson Education India,
3. Sinha, Sanjay Kumar. "**CNC programming using FANUC custom macro"** McGraw Hill Professional.
4. Luggen, William W. "**Flexible manufacturing cells and systems**" Prentice Hall.

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B. Tech. Mechanical Engineering

Semester – VII

24ME439T					Non-Destructive testing and failure analysis					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To introduce and familiarize the different Non-destructive testing techniques and its importance in manufacturing.
2. To understand basic working principle and its allied aspects for different widely used NDT techniques.
3. To enable the students for doing Failure analysis of a component and make them aware about different tools used.
4. To make student aware about advanced tools pertaining to Industry 4.0 in NDT.

UNIT 1 FAILURE ANALYSIS 10 Hrs.

Failure analysis – methodology; approaches, tools and techniques of failure analysis; modes of failure; failure data retrieval; procedural steps for Investigation of a failure for failure analysis. **Techniques for Failure identification:** Optical Microscopy, X-RAY Diffraction, Electron microscopy

UNIT 2 INTRODUCTION TO NDT 10Hrs.

Introduction to Non-Destructive Testing (NDT), its relevance, importance and key features. Classification of NDT techniques and its various applications. Selection of NDT techniques and their importance in the field of reliability analysis as well as remaining life assessment (RLA). Brief introduction to 16 methods of NDT and brief introduction to 6 widely used NDT techniques. **Visual examination:** Fundamentals of defects, root causes of defects, tools and techniques for visual observation. Acceptance standards and application. Liquid penetrant testing – procedure; penetrant testing materials, penetrant testing method – sensitivity; application and limitations.

UNIT 3 MAGNETIC PARTICLE TESTING 10 Hrs.

MPT: Definition and principle; magnetizing technique, procedure, equipment sensitivity and limitation; Application & Acceptance Standards **RADIOGRAPHY:** Basic principle, electromagnetic radiation in film, radiographic imaging, inspection techniques, applications, limitations, real time radiography, safety in Industrial radiography. Application & Acceptance Standards

UNIT 4 ADVANCED TECHNIQUES 10 Hrs.

Ultrasonic Techniques: Ultrasonic transducers, inspection methods, technique for normal beam inspection, flaw characterization technique, ultrasonic flaw detection equipment modes of display, immersion testing, advantage, limitations; Application & Acceptance Standards **Eddy current testing:** Principle, instrument techniques, sensitivity application, limitation; ultrasonic testing – basic properties of sound beam, Application & Acceptance Standards **Acoustic emission testing:** Principles of AET and techniques, its application, Acceptance standards. Use of automation, AI and Industry 4.0 in testing techniques and its case studies

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Understand the basic concept of Failure analysis and apply it for remaining life assessment.
CO2 : Evaluate the basis of NDT and principles of visual testing.
CO3 : Explain the various methods of NDT such as, magnetic particle testing and liquid penetrant testing.
CO4 : Demonstrate the NDT method of RT and UT and its application on real life manufacturing scenario.
CO5 : Evaluate the advanced NDT technique such as AET and ECT for manufacturing applications.
CO6 : Compose the industry standards of NDT and use of industry 4.0 tools for Non-destructive testing.

TEXT/REFERENCE BOOKS

1. Baldev Raj, Jayakumar T., Thavasimuthu M., 'Practical Non-Destructive, 2002
2. Testing', Narosa Publishing, 1997.
3. Das A.K., 'Metallurgy of Failure Analysis', TMH, 1992.
4. Hull., 'Non-Destructive Testing', ELBS Edition, 1991
5. Halmshaw R., - 'Non-Destructive Testing', Edward Arnold, 1990

Pandit Deendayal Energy University, School of Technology

B. Tech. Mechanical Engineering

Semester – VII

24ME440T					Material and Procurement Management					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	0	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To understand the concept of material management and its application.
2. To introduce the Capacity Management and Planning and understand its application.
3. To study the Demand Management and Forecasting along with their applications.
4. To study applications of Economic-Order Quantity and Period-Order Quantity.

UNIT I: INTRODUCTION TO MATERIAL MANAGEMENT 10 Hrs.

Introduction, Sales and Operations Planning, Manufacturing Resource Planning, Enterprise Resource Planning, Making the Production Plan, Master Scheduling, Developing a Master Production Schedule , Production Planning, Master Scheduling, Material Requirements Planning, Bill of Material, Material Requirements Planning Process, Using the Material Requirements Plan.

UNIT II: CAPACITY MANAGEMENTAND PURCHASING 10 Hrs.

Capacity Management and Planning, Capacity Requirements Planning (CRP), Capacity Available, Capacity Required (Load), Scheduling Orders, Making the Plan, Production Activity Control, Data Requirements, Order Preparation, Load Levelling, Scheduling Bottlenecks, Theory of Constraints and Drum-Buffer-Rope, Implementation, Control, Production Reporting. Purchasing: Establishing Specifications, Functional Specification Description, Selecting Suppliers, Price Determination, Impact of Material Requirements Planning on Purchasing , Expansion of Purchasing in to Supply Chain, Management, Some Organizational Implications of Supply Chain, Management.

UNIT III: FORECASTING 10 Hrs.

Demand Management, Demand Forecasting, Characteristics of Demand, Principles of Forecasting, Collection and Preparation of Data, Forecasting Techniques, Some Important Intrinsic Techniques, Seasonality, Tracking the Forecast. Inventory Fundamentals, Aggregate Inventory Management, Item Inventory Management, Inventory and the Flow of Material, Supply and Demand Patterns, Functions of Inventories, Objectives of Inventory Management, Inventory Costs, Financial Statements and Inventory, ABC Inventory Control

UNIT IV: ORDER QUANTITIES 10 Hrs.

Economic-Order Quantity (EOQ), Variations of the EOQ Model, Quantity Discounts, Order Quantities for Families of Product When Costs Are Not Known, Period-Order Quantity (POQ), Independent Demand Ordering Systems, Order Point System, Determining Safety Stock, Determining Service Levels, Different Forecast and Lead-Time Intervals, Determining When the Order Point Is Reached, Periodic Review System, Distribution Inventory

TOTAL HOURS: 40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Identify the role of supply chain management in balancing the conflicting objectives of marketing, finance and production
CO2 : Integrate the sales, production planning and master production schedule to develop an overall plan.
CO3 : Apply various purchasing method and inventory controlling techniques into practice.
CO4 : Analyse the impact of material requirements planning and capacity management on purchasing.
CO5 : Evaluate various facility alternatives and their capacity decisions.
CO6 : Develop a balanced line of production that support organizations strategic goals.

TEXT/REFERENCE BOOKS

1. J. R. Tony Arnold, Stephen N. Chapman, Lloyad M.Clive, **"Introduction to Material Management"**, PHI Learning.
2. Datta, A.K., **"Material Management Procedure: Text and Cases"**, PHI Learning.
3. Dobbler, B., **"Purchasing and Supply Management"**, Tata McGraw Hill.
4. Gopalakrisnana, P. and Sundaraesan M., **"Material Management: An Integral Approach"**, PHI Learning.
5. Varma, M.M., **"Material Management"**, Sultan Chand and Sons.
6. Menon, K. S., **"Purchasing and Inventory Control"**, Wheeler Publishers

Pandit Deendayal Energy University, School of Technology

B. Tech. Mechanical Engineering

Semester – VII

24ME440T					Material and Procurement Management					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	0	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To understand the concept of material management and its application.
2. To introduce the Capacity Management and Planning and understand its application.
3. To study the Demand Management and Forecasting along with their applications.
4. To study applications of Economic-Order Quantity and Period-Order Quantity.

UNIT I: INTRODUCTION TO MATERIAL MANAGEMENT 10 Hrs.

Introduction, Sales and Operations Planning, Manufacturing Resource Planning, Enterprise Resource Planning, Making the Production Plan, Master Scheduling, Developing a Master Production Schedule , Production Planning, Master Scheduling, Material Requirements Planning, Bill of Material, Material Requirements Planning Process, Using the Material Requirements Plan.

UNIT II: CAPACITY MANAGEMENTAND PURCHASING 10 Hrs.

Capacity Management and Planning, Capacity Requirements Planning (CRP), Capacity Available, Capacity Required (Load), Scheduling Orders, Making the Plan, Production Activity Control, Data Requirements, Order Preparation, Load Levelling, Scheduling Bottlenecks, Theory of Constraints and Drum-Buffer-Rope, Implementation, Control, Production Reporting. Purchasing: Establishing Specifications, Functional Specification Description, Selecting Suppliers, Price Determination, Impact of Material Requirements Planning on Purchasing , Expansion of Purchasing in to Supply Chain, Management, Some Organizational Implications of Supply Chain, Management.

UNIT III: FORECASTING 10 Hrs.

Demand Management, Demand Forecasting, Characteristics of Demand, Principles of Forecasting, Collection and Preparation of Data, Forecasting Techniques, Some Important Intrinsic Techniques, Seasonality, Tracking the Forecast. Inventory Fundamentals, Aggregate Inventory Management, Item Inventory Management, Inventory and the Flow of Material, Supply and Demand Patterns, Functions of Inventories, Objectives of Inventory Management, Inventory Costs, Financial Statements and Inventory, ABC Inventory Control

UNIT IV: ORDER QUANTITIES 10 Hrs.

Economic-Order Quantity (EOQ), Variations of the EOQ Model, Quantity Discounts, Order Quantities for Families of Product When Costs Are Not Known, Period-Order Quantity (POQ), Independent Demand Ordering Systems, Order Point System, Determining Safety Stock, Determining Service Levels, Different Forecast and Lead-Time Intervals, Determining When the Order Point Is Reached, Periodic Review System, Distribution Inventory

TOTAL HOURS: 40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 : Identify the role of supply chain management in balancing the conflicting objectives of marketing, finance and production
CO2 : Integrate the sales, production planning and master production schedule to develop an overall plan.
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CO6 : Develop a balanced line of production that support organizations strategic goals.

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2. Datta, A.K., **"Material Management Procedure: Text and Cases"**, PHI Learning.
3. Dobbler, B., **"Purchasing and Supply Management"**, Tata McGraw Hill.
4. Gopalakrisnana, P. and Sundaraesan M., **"Material Management: An Integral Approach"**, PHI Learning.
5. Varma, M.M., **"Material Management"**, Sultan Chand and Sons.
6. Menon, K. S., **"Purchasing and Inventory Control"**, Wheeler Publishers

Pandit Deendayal Energy University, School of Technology

B. Tech. Mechanical Engineering

Semester – VII

24ME401T					Optimization Techniques					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	1	0	4	4	25	50	25	--	--	100

COURSE OBJECTIVES

1. To impart the knowledge of unconstraint optimization techniques
2. To learn constraints optimization techniques
3. To understand classical and non-traditional optimization techniques

UNIT I: PROBLEM FORMULATION AND GRAPHICAL SOLUTIONS 10Hrs.

Introduction, need of optimization, objective function, constraints, design variables, feasible and infeasible solutions, classification of optimization problems, applications. Graphical Method: different cases, Feasible and infeasible solutions, redundant constraints, unbound solutions, unique solution.

UNIT II: LINEAR PROGRAMING PROBLEM AND SOLUTIONS 12 Hrs.

Linear Programming, Problem formulation, simplex method, Big-M method for LPP, transportation and assignment problems, Duality in Linear Programming

UNIT III: SINGLE VARIABLE PROBLEM AND SOLUTIONS 12 Hrs.

Unconstrained optimization techniques, elimination methods, interpolation methods, direct and indirect search methods

UNIT IV: 4 MULTI VARIABLE PROBLEM AND SOLUTIONS 11 Hrs.

Classical optimization techniques, single variable, multivariable optimization techniques, random search methods, methods of feasible directions, Genetic Algorithm, Simulated Annealing, Teaching-learning-based optimization

TOTAL HOURS: 45 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1	:	Understand basic concepts of mathematics to formulate an optimization problem and solve it graphically.
CO2	:	Use LP techniques to obtain the solution of the simplex and dual simplex problem.
CO3	:	Evaluate various assignment problem
CO4	:	Evaluate various transportation problem
CO5	:	Implement direct and indirect search method for single variable optimization problem
CO6	:	Apply search method for multi-variable optimization problem

TEXT/REFERENCE BOOKS

1. S S Rao, "**Engineering Optimization**", New Age International
2. K Deb, "**Optimization for Engineering Design: Algorithms & Examples**", Prentice-Hall of India
3. C Pearce, E Hunt , "**Optimization: Structure and Applications**", Springer
4. Ravindran, G. Reklaitis, K. M. Ragsdell, "**Engineering Optimization: Methods and Applications**", Wiley
5. F Hillier, G Lieberman, "**Introduction to Operation Research**", Mc Graw Hill
6. H Taha, **Operation Research** –, Pearson Education.
7. A Verma, "**Operation Research**", S.K. Kataria and Sons.
8. V Kapoor, "**Operation Research**", Sultan Chand & Sons

Pandit Deendayal Energy University, School of Technology

B. Tech. Mechanical Engineering

Semester – VII

24ME402T					Project Management					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	1	0	4	4	25	50	25	--	--	100

COURSE OBJECTIVES

1. To understand the importance of project Management in manufacturing and service sectors
2. To define project requirements and identify stakeholders
3. To create project plan and schedule for monitoring project deliverables.
4. To deliver successful projects that support organizations strategic goals.

UNIT I: INTRODUCTION TO PROJECT MANAGEMENT 10Hrs.

Justifying Project Management, Projects – Definitions, The Project Management Triangle – Scope, Time and Cost, What is Project Management, Projects & Operations, The Project Life Cycle, Project Stakeholders, Project Management Process, Groups, Project Management Knowledge Areas, Project Initiation: The Project Charter, The Project Scope Document, Work Breakdown
Structures, Project Contracts – Scope, Delivery, Costs and Risks.

UNIT II: PROJECT PLANNING AND SCHEDULING 10 Hrs.

Project Network Representations, Activity Parameter, Estimation – Time, Cost and Resources, Project Time Schedule, Gantt Charts, CPM and PERT, Activity and Project Crashing, Resources Scheduling

UNIT III: PROJECT EXECUTION MANAGEMENT 10 Hrs.

Quality Specifications, Quality Control Tools, Resources Procurement and Allocation, Systems and Processes, Communications and Documentation, Managing Teams, Resources Demobilization, Project Simulation and Risk Assessment, Use of IT tools.

UNIT IV: PROJECT MONITORING & CONTROL PROCESS 10 Hrs.

Monitoring Process (on a regular basis)-Monitoring and controlling activities, Key Results from monitoring process; Techniques to evaluate project performance- Earned Value Analysis, The Critical Ratio, Review Meetings; Project Crashing and Time-Cost Trade-Off**;** Project Closure and Review**:** Performance Evaluation – Scope, Time and Cost, Performance of
Teams, Lessons Learnt, Project Closure Report. Integrated Examples/Cases.

TOTAL HOURS: 40 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1: Identify the project requirements and stakeholders.

CO2: Understand the importance of project Management in manufacturing and service sectors.

CO3: Apply the methods, activities and risks involved in projects.

CO4: Create project plan and schedule for monitoring project deliverables.

CO5: Develop necessary skills to manage stakeholders by IT tools.

CO6: Deliver successful projects that support organizations strategic goals.

TEXT/REFERENCE BOOKS

1. Mantel Jr., Samuel J., Jack R. Meredith, Scott M. Shafer, Margaret M. Sutton with M. R. Gopalan, "**Project Management Core Text Book**", John Wiley & Sons.
2. Meredith, Jack R., and Samuel J. Mantel, Jr., **"Project Management: A Managerial Approach**", John Wiley & Sons.
3. Pinto, Jeffrey K. **"Project Management: Achieving Competitive Advantage and MS Project**", Pearson Publishing.
4. Maylor Harvey, **"Project Management**", Pearson Publishing.

Pandit Deendayal Energy University, School of Technology
B. Tech. Mechanical Engineering
Semester – VII

24ME403P					Computational Engineering Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	4	2	04	-	-	-	25	25	50

COURSE OBJECTIVES

1. Learn computational Software for numerical calculations.
2. Understand the basic commands of the computational software.
3. Apply the software programming to solve real life problems.

Module-1 Introduction to Numerical Methods and computational applications

a. Introduction to the Numerical methods, Discretization of domain and equations, errors.
b. Introduction to computational software: Matrices and Basic Graph plotting, Data Files and Text Strings, Basic Programming and User defined functions and function files

Module-2 Numerical Methods for Mechanical Engineering problems

a. Mathematical modelling of an Engineering System and numerical Solution to ODE
b. Mathematical modelling of an Engineering System and numerical Solution to PDE
c. Solution of Simultaneous Algebraic Equations and numerical tools for Eigen value problem.
d. Solution of Algebraic and Transcendental Equations
e. Numerical Method for Interpolations and Regression analysis
f. Numerical method for Differentiation and Integration
g. Demonstration of advanced packages or open-ware toolkit based using the software

Module-3 Case-Studies and mini-projects from Mechanical engineering to demonstrate the acquired numerical skills

a. Case Study 1
b. Case Study 2

Max hours: 52

COURSE OUTCOMES

On completion of the course, student will be able to

CO-1: **Understand** the features of the software program.
CO-2: **Apply** basic operations of the computational software and **plot** and **analyse** results.
CO-3: **Solve** the ODE using numerical methods and software tools.
CO-4: **Solve** the PDE using numerical methods and software tools.
CO-5: **Understand and apply** different numerical methods and **develop** programs based on respective numerical methods.
CO-6: **Apply** the software programming to real applications/Engineering case-studies.

TEXT/REFERENCE BOOKS

1. Chapra, Steven, Applied Numerical Methods with MATLAB for Engineers and Scientists. McGraw Hill, 2011.
2. B. S. Grewal, Numerical Methods in Engineering & Science with Programs in C, C++ & MATLAB. Khanna Publishers 10th edition, 2012.
3. Xu, Sheng. An Introduction to Scientific Computing with MATLAB® and Python Tutorials. CRC Press, 2022.